

SAN ANTONIO INTL, TX, USA

WMO: 722530

Lat: 29.544N

Lon: 98.484W

Elev: 241

StdP: 98.47

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12921

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.1	0.6	-11.5	1.4	5.8	-9.0	1.8	8.4	10.8	14.3	9.3	12.8	3.4	10	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.9	37.4	23.1	36.5	23.1	35.5	23.2	25.6	31.1	25.2	30.7	24.9	30.2	4.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.4	19.9	26.8	24.0	19.4	26.7	23.6	19.0	26.6	79.7	31.2	78.1	30.6	76.8	30.1	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.5	7.8	-4.4	39.4	1.8	1.5	-5.7	40.5	-6.8	41.4	-7.8	42.2	-9.2	43.3		
			-6.0	26.4	1.7	0.6	-7.2	26.8	-8.1	27.1	-9.1	27.5	-10.3	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.3	11.6	14.0	17.3	21.2	25.2	28.3	29.5	29.9	26.9	22.0	16.5	12.3	
	DBStd	7.58	5.00	5.47	4.73	3.80	3.01	2.02	1.70	1.82	2.87	4.26	4.90	4.96	
	HDD10.0	109	41	22	6	1	0	0	0	0	0	1	7	32	
	HDD18.3	751	213	140	76	17	1	0	0	0	1	17	90	197	
	CDD10.0	4217	91	133	232	335	470	549	604	617	506	374	203	103	
	CDD18.3	1817	5	17	44	102	213	299	346	359	257	132	36	9	
	CDH23.3	19201	32	116	269	769	1924	3369	4172	4499	2692	1111	205	41	
	CDH26.7	8928	2	26	56	236	777	1624	2159	2418	1244	356	29	2	
Wind	WSAvg	3.8	3.4	3.7	4.1	4.4	4.5	4.3	4.0	3.7	3.2	3.4	3.4	3.3	
Precipitation	PrecAvg	808	43	47	45	61	104	98	57	65	83	94	66	46	
	PrecMax	1328	216	163	155	224	326	304	430	283	332	459	240	355	
	PrecMin	417	0	0	1	0	3	0	0	0	1	3	0	2	
	PrecStd	226	42	37	34	48	79	75	84	64	71	88	56	59	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.2	29.7	30.4	33.9	35.8	38.0	38.3	38.6	37.7	33.4	29.3	26.7	
		MCWB	15.7	16.2	18.5	20.6	23.0	23.2	22.6	23.0	22.2	22.2	20.9	18.2	
	2%	DB	23.9	26.5	28.1	31.2	34.0	36.2	37.1	37.5	35.6	31.7	27.3	24.4	
		MCWB	15.4	16.8	18.4	20.3	22.8	23.3	23.0	23.3	22.7	22.2	19.7	17.4	
	5%	DB	21.9	24.3	26.4	29.2	32.4	34.9	36.1	36.6	34.1	30.3	25.7	22.6	
		MCWB	15.5	16.7	18.2	20.2	22.7	23.4	23.2	23.2	22.7	21.6	19.1	17.0	
	10%	DB	20.2	22.2	24.4	27.6	30.9	33.5	34.9	35.5	32.6	28.7	24.1	20.8	
		MCWB	15.1	16.2	17.7	19.8	22.5	23.4	23.4	23.4	22.6	21.0	18.6	16.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.2	20.7	21.8	24.0	25.7	26.0	25.8	25.8	25.6	25.1	23.0	21.2
			MCDB	21.9	24.0	26.0	29.0	31.4	31.8	31.2	31.9	30.2	28.8	26.2	23.8
2%		WB	19.1	19.9	20.9	23.0	24.9	25.5	25.4	25.3	25.1	24.4	22.1	20.2	
		MCDB	21.3	22.9	24.3	27.5	30.3	31.2	31.0	31.0	29.5	27.9	24.9	22.3	
5%		WB	17.9	19.1	20.3	22.3	24.2	25.0	25.1	25.0	24.6	23.8	21.2	19.1	
		MCDB	20.4	21.9	23.5	26.3	29.0	30.6	30.5	30.7	29.1	27.2	24.0	21.3	
10%		WB	16.4	18.2	19.6	21.6	23.7	24.6	24.7	24.6	24.2	23.1	20.2	17.5	
		MCDB	19.1	20.9	22.7	25.3	28.3	29.8	29.9	30.2	28.6	26.4	23.1	20.0	
Mean Daily Temperature Range	5% DB	MDBR	11.8	11.6	11.4	11.4	10.5	10.3	10.5	10.9	10.7	11.1	11.2	11.5	
		MCDBR	14.2	13.9	12.8	12.8	11.9	11.9	12.1	12.2	12.2	11.2	11.7	13.3	
		MCWBR	8.3	7.5	6.2	5.6	3.8	2.8	2.6	2.6	3.2	4.3	6.1	7.9	
	5% WB	MCDBR	10.2	10.1	9.9	9.7	9.6	9.6	9.9	10.6	9.6	8.8	9.2	10.1	
		MCWBR	7.5	6.9	5.9	5.1	4.0	3.1	2.6	2.7	3.2	4.3	5.9	7.7	
Clear-Sky Solar Irradiance	taub		0.326	0.332	0.356	0.399	0.444	0.428	0.436	0.421	0.414	0.373	0.360	0.318	
	taud		2.502	2.487	2.433	2.330	2.249	2.346	2.360	2.382	2.355	2.458	2.427	2.536	
	Ebn at Noon		913	936	931	896	852	861	853	865	861	880	863	905	
	Edh at Noon		90	99	112	128	140	126	124	120	120	101	97	83	
All-Sky Solar Radiation	RadAvg		3.02	3.55	4.49	5.33	5.75	6.60	6.65	6.36	5.21	4.31	3.24	2.68	
	RadStd		0.34	0.64	0.46	0.53	0.60	0.43	0.58	0.37	0.52	0.57	0.36	0.34	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+95	+62
(47) Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+86

Nomenclature: See separate page